Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/25/2008
PI Name:	Thomas, John Ph.D.		
Project Title:	Quantum Coherence in Ultracold Fermionic Vapors		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27708	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2001	**End Date:**	04/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNCO4GA14G		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

The purpose of this program is a fundamental study of coherence between ground state hyperfine levels in an ultracold vapor of lithium-6 fermions contained in an ultrastable CO2 laser trap. This system permits important proof-of-principle experiments which exploit the nointeracting properties of a single component Fermi gas.

The objective of this program is to broadly investigate coherence between ground hyperfine states in an ultracold fermionic gas of lithium-6 contained in an ultrastable all-optical trap. For fermions, the Pauli exclusion principle provides unique opportunities for increasing the spectroscopic resolultion. Proposed experiments include measurement of collisional shifts to determine the magnitude and sign of elastic scattering amplitudes in two-component mixtures, demonstration of double suppression of shifts and inelastic rates for coherent superposition states, demonstration of super-long coherence lifetimes for superposition states fo fermions in an ultrastable optical trap, and potentially

Task Description:	measurement of frequency shifts arising from Cooper pairing. The ultimate coherence lifetime is expected to be limited by tthe gravitational potential and the Pauli exclusion principle. The significance of the proposed program is two-fold. First, the program explores new methods for suppressing interactions by using fermionic atoms. This may impact the development of sensitive high resolution spectroscopic methods for precision measurement and atomic clocks. Further, the program utilizes and ultrastable all-optical trap with an exceptionally long lifetime and extremely low residual heating rates. This system will enable storage and study of diamagnetic atoms and molecules which could not otherwise be trapped. Second, the lithium-6 system exhibits magnetically tunable pairing interactions and is therefore an ultracold gas analog of a superconductor. Hence, it will enable precision tests of superconductivity theories in systems with widely tunable interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research on strongly-interacting Fermi gases impacts the theory of super-high temperature superconductors, which one day may enable lossless communication, power distribution, and magnetic levitators for transportation.
Task Progress:	Our focus in the current year has been on measurement of the entropy and sound velocity in a strongly-interacting Fermi gas. The entropy of such a gas is of great theoretical interest, and is of particular importance for exploring theories of high temperature superfluidity and superconductivity. We have constructed a new second apparatus for our our entropy measurements, which are accomplished by combining two techniques, an adiabatic sweep of the magnetic field, and the use of the virial theorem for energy measurement. In September of 2005, we published a paper in PRL which showed that the energy of a harmonically trapped, strongly-interacting Fermi gas is proportional to the mean square cloud size, in exactly the same way as that of an ideal noninteracting Fermi gas. This is a consequence of universal behavior, and arises when the zero energy scattering length is infinite, so that the particle density sets the only length scale. Then, the local pressure is a function only of the local density and of the temperature, and the theorem follows. To measure the entropy, we begin by creating a strongly interacting Fermi gas of 6Li at at a magnetic field of 834 G, near a Feshbach resonance. We cool the gas to very low temperature by evaporative cooling in our optical trap, producing essentially the ground state, which is a superfluid. Then we add energy to the gas by releasing the atomic cloud from our optical trap and then recapturing it after a short time. The kinetic and interaction energy are conserved, but the potential energy increases when the trap is reinstated onto the expanded cloud. The size of the cloud is measured. Then the same sequence is done again, but this time the magnetic field is swept to 1200 G after the energy is added. Then the cloud size is measured. At 1200 G, the gas is very weakly interacting, and the entropy is well known, and determined from the energy, which is measured from the cloud size. Since the sweep is adiabatic, as verified by doing a round trip, the entropy of the strongly interacting gas is the same as that of the weakly interacting gas at 1200 G. We are also completing measurements of sound velocity on our older apparatus. We create a strongly-interacting Fermi gas in the form of a cigar-shaped cloud, about 200 mm. We excite a density perturbation by applying a pulse of blue detuned light, which is focused onto one end of the cloud. We measure the propagation of the perturbation across the cloud over a time period of 10 ms. By plotting the position versus time, we precisely extract the sound velocity. These measurements are done over a wide range of magnetic fields, permitting studies of superfluids ranging from Bose-Einstein condensates of diatomic molecules at low magnetic fields to strongly and weakly interacting Fermi superfluids at higher fields. The measurements test the best quantum many-body calculational methods in the nonperturbative regime, and appear to support current quantum Monte Carlo predictions of the equation of state over a wide range of interaction strengths. Supplemental (10/1/2006-4/30/07) In the current year, we achieved two major breakthroughs. We completed the first model-independent measurements of the energy and entropy in a strongly-interacting Fermi gas. The entropy of such a gas is of great theoretical interest, and our data now provides the first precision benchmark for testing the best current nonperturbative calculational methods. These include both numerical and analytical methods, which apply to theories of high temperature superfluidity and superconductivity. We also completed the first measurements of the sound velocity in a Fermi gas near a Feshbach resonance, as a function of the interaction strength. Our measurements test the zero temperature equation of state. Presentations (Sept,., 2006-April, 2007) Thomas, J., ``Optically-trapped, strongly interacting Fermi gases," The College of William and Mary (September 22, 2006, Williamsburg, VA).  Thomas, J. ``Optical trapping of atomic Fermi gases," Fitzpatrick Center, Duke University (November 7, 2006, Durham, NC). Thomas, J., ``Optically-trapped, strongly interacting Fermi gases," University of Maryland (November 14, 2006, Baltimore, Md.).  Thomas, J., ``Quantum viscosity of a strongly-interacting Fermi gas," Workshop ``The interface of quark-gluon plasma physics and cold-atom physics," (Trento, Italy, March 19-23, 2007). Talk given by Andrey Turlapov.  ``Quantum viscosity of a strongly-interacting Fermi gas," Workshop ``The interface of quark-gluon plasma physics and cold-atom physics," (Trento, Italy, March 19-23, 2007). Talk given by Andrey Turlapov.
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